

Design and analysis of smooth geometry-conforming lattices via Generalized Bézier patches

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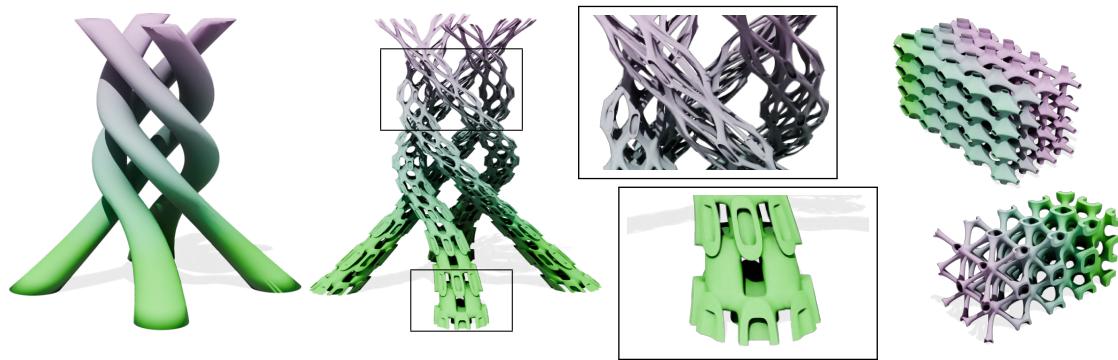


Figure 1: A solid model (left) is filled by heterogeneous and globally G^1 smooth metamaterial (middle) using Generalized Bézier surfaces. The tiles allow for graded shape and thickness throughout the volume, thus generating non-uniform mechanical/physical behaviour in the filling material. The geometry (and color/material) of the metamaterial can also vary according to the topology of the canonical tile used (right), and be optimized using isogeometric analysis tools.

Abstract

We propose a novel methodology to model globally G^1 geometry-conforming metamaterials using Generalized Bézier-based (GB) patches. Since every canonical tile is inscribed in a unit cube, we exploit its natural decomposition in octants and design multi-sided patches for each octant. The orthogonality constraints in each octant's GB surface ensure global G^1 continuity in the entire parametric domain and, under the assumption of a global smooth mapping that fills a desired free-form volume by deformed cubes, results in globally G^1 metamaterial in physical space. The presented tiles can be graded in both shape and thickness using only a few shape parameters preserving their continuity properties, thus enabling optimization of novel metamaterials in various applications. Using isogeometric analysis tools we show that a G^1 metamaterial results in a smoother stress field than a typical G^0 tile used in state-of-the-art parametric approaches. We further show that our tile, given the continuity constraints, is lighter in terms of necessary control points than state-of-the-art parametric tiles and can approximate popular implicit tiles with sufficient accuracy. Finally, our pipeline is demonstrated in several design/analysis examples.

CCS Concepts

• **Applied computing** → **Computer-aided design**; • **Computing methodologies** → **Volumetric models**; **Shape analysis**;

1. Introduction

With the growing sophistication of 3D printing technologies, it is now possible to fabricate objects with non-uniform material distributions that use less material to achieve a given shape while outperforming their solid counterparts (e.g., by being lighter). These objects are made by filling their volume with repetitions of one or several canonical shapes called *tiles*. The nature, properties and ways of disposition of these tiles is the subject of *metamaterials* (also called *microstructures* or *lattices* in some communities) research [TKZ15,NV16]. One important question is how the volume-filling canonical tiles interact with the border of the object's shape. In this sense, metamaterials can either be *geometry-conforming* or not, i.e. they can either adapt to the shape of the object via deformation or via clipping. We are interested in geometry-conforming metamaterials, see Fig. 2. The usual case made for geometry-conforming metamaterials is that they present better mechanical properties than non-conforming ones that clip tiles on the boundary, a critical zone in any physical part. In contrast, in the case of geometry-conforming metamaterials, no cell is ever clipped, just deformed, thus preserving better the topological/physical/mechanical behavior of the original tile [ZL25].

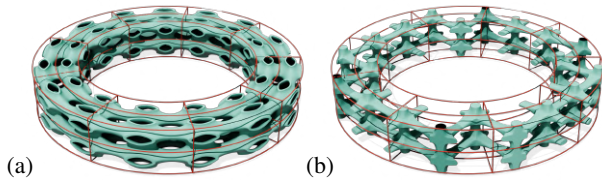


Figure 2: Geometry conforming vs. non-conforming approaches to lattice design. (a) Conforming structures are achieved via deformation while (b) non-conforming ones via clipping.

This work proposes a new methodology for the design of geometry-conforming metamaterials based on Generalized Bézier-based (GB) patches. To design the so called *canonical tile* (a tile embedded in a unit cube in the parametric space), we exploit the natural cube's symmetry and consider only the part contained in one octant of the cube. Such a patch is, for most of the state-of-the-art tiles [ZL25,Elb23], an n -sided surface, typically $n = 6$. This hints that an efficient way to represent tiles are n -sided parametric patches [VSK16]. Also, since we are looking for a smooth tile, n -sided patches provide a naturally smooth surface in each octant that is difficult to achieve using unions of four-sided patches.

To design a canonical tile that is G^1 , aka tangent, continuous, we incorporate orthogonality constraints along the patch boundaries, which can be done via the correct disposition of the control point structure and, using a smooth global mapping, this construction propagates via functional composition into a globally G^1 design of the whole 3D free-form object. We further show that the proposed design pipeline also directly leads to volumetric representations (V-reps) [ME16], and is sufficiently flexible to design tiles of graded shapes and thickness, which is an important feature when optimizing physical properties of the material as a whole.

Design is, however, only part of the task for today's demand on efficient design-analysis integration. Trivariate-based tiles are especially well suited for these demands as they are analysis-ready

in an isogeometric analysis (IGA) setting [HCB05,CHB09,BH22]. This tight integration enables efficient feedback loops between design and analysis, thus enabling the optimization of lattice designs towards targeted mechanical properties and kinematic responses. In our work, we show a way to make our novel multi-sided patch design compatible with state-of-the-art IGA tools.

1.1. Overview and contributions

We summarize our contributions as:

- Using multi-sided generalized Bézier patches, we propose a new modeling pipeline to design G^1 -continuous canonical tiles (Sec. 3.2) and their dual variants (Sec. 3.3).
- We show that this tile construction is easily compatible with functional composition and design a variety of geometry-conforming lattices using only a few parameters per tile, yet with sufficient flexibility to generate heterogeneous lattices of variable shape and thickness (Sec. 5).
- Demonstrate the compatibility with the IGA tools and show that the proposed framework can be used to tune the lattice geometry for potential mechanical applications (Sec. 6).

2. Related work

To generate a geometry-conforming lattice of a 3D (solid) object one needs a volumetric parameterization of the interior of the volume, i.e. a map to guide the disposition of the tiles. This is no trivial matter, since most computer-aided design methodologies have evolved following a surface-based design paradigm, only until very recently shifting to volumetric representations [Elb16,Elb23]. Additionally, one wants this parameterization to be *as smooth as possible* to avoid virtual internal discontinuities which are not really present in the physical object. In this sense, approaches to lattice metamaterial design can be classified according to the nature of the parameterization used: some works use discretizations of the volume such as voxels or hexahedral meshes [TDZZ17,BWWW25], but these methods work, precisely because of this choice, only with discrete approximations of the true shape of the object; other works use globally smooth volumetric parameterizations such as trivariate Bézier or B-Spline solids which describe exactly the true shape of the object [Elb16,Elb23].

In the trivariate-based methods, once a parameterization of the volume is given, the lattice is built by populating the parametric space with repetitions of the canonical tiles and, via the mapping, deforming them to fill the target volume in physical space. This is known as *spline composition*, see [DGHM93,Elb16] and Fig. 3. Notice that, for the goal of creating globally G^1 structures in physical space, having a sufficiently smooth volumetric parameterization reduces the challenge to creating globally G^1 structures in parametric space, since the mapping will preserve the continuity. Further down the line, the matter comes to the nature (i.e. geometric representation) and properties (e.g. continuity) of the tile used. One option to have parametric tiles is to build them using trivariate solids themselves or B-Spline surfaces [Elb23], but creating globally G^1 multi-patch geometries in this setting is not an easy task. We propose to use a new kind of surfaces that generalize Bézier surfaces, known as Generalized Bézier surfaces [VSK16], which are defined

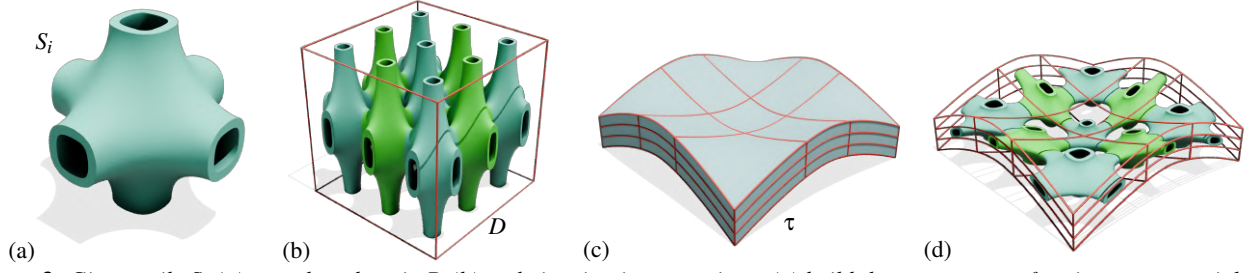


Figure 3: Given a tile S_i (a), populate domain D (b) and via trivariate mapping τ (c) build the geometry-conforming metamaterial (d). In our pipeline, the canonical tile S_i can be a surface, a shell (as shown in (a)), or a fully volumetric object.

as combinations of boundary-based local ruled surfaces that ensure G^1 control over multi-sided patches but are also apt for the spline composition approach, thus taking advantage of a well-established design philosophy.

In addition to the smoothness constraints, usual applications of lattice metamaterials involve a large number of cells. Because of this, representation compactness is paramount if one desires a method to be useful for designers and engineers [ZL25]. In this regard, parametric tiles, being fully described only by their control points, are a powerful tool. In this work, we intend to show that the use of Generalized Bézier-based tiles reduces the number of control points required to describe the tile and enables a wider range of shapes with just a couple of geometric parameters.

Our framework uses the so called *multi-sided* (aka *n-sided*) Generalized Bézier patches [VSK16]. These surfaces are generalizations of the classical tensor-product Bézier patches with four sides (and vertices) towards larger number of sides. These parametric patches have similar representation and are easy to control via a control point network, yet the domain is not a rectangle, but an n -gon. The n -sided surfaces can be also represented implicitly [SVSV20a], however, then one has to deal with trimming planes and therefore we opt for the parametric variant. For more papers on n -sided patches, the reader is referred to the recent survey paper [VSV24]. Implicit tiles have been considered recently too [HAEK24] and even though the modeling side is elegant, to deal with optimization, one has to trim the domain which is non-trivial.

To numerically simulate the behavior of lattices, specific modeling and numerical methods are required. The computational cost in terms of memory and time is indeed very demanding, or even intractable, if standard approaches such as the finite element method are used as black-box solvers. The most common trend in analyzing lattices is therefore to rely on multiscale approaches [SJ22, GFW21, MKBM23]. In other words, the unit cells (or tiles) are seen here as materials: the full fine-scale geometrical details are replaced by specific constitutive relations reflecting the (average) physical behaviors of the cells. Such multiscale-based strategies have been successfully applied in optimization frameworks, particularly in topology optimization [AJM04a, HTK13]. In this context, they enable the identification of unit cells with desired effective mechanical properties, such as a targeted one-dimensional macroscopic stress-strain relationship [BK23, TED*25]. In addition, multiscale-based design optimization frameworks may facili-

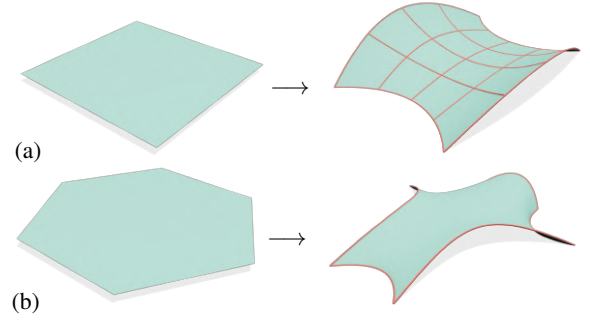


Figure 4: A standard (four-sided) tensor product Bézier patch (a) and the n -sided generalization (b). The rectangular parametric domain of the Bézier patch generalizes to a convex polygon with n sides; here $n = 6$.

tate the distribution and orientation of pre-selected and pre-studied unit cells within a macro-domain to achieve an optimal lattice structure [AGDP19, CSS22, LGXZ22, FT21]. Other relevant research deals with interior designs that mitigate large deformations via shock-absorbing microstructures [HPZ23].

However, it should be noted that multiscale methods require a clear scale separation. This assumption is questionable in the case of lattice metamaterials, since 3D printers constrain the achievable length-scale range of cells, and the macro-shape often follows a rather slender geometry, hence with few cells in one direction [DSRRL*25]. As an alternative to multiscale-based approaches, it is possible to employ dedicated solvers that take advantage of the quasi-periodicity within lattices and simulate the full fine-scale problems while using limited computational resources. These solvers may involve fast formation of finite element operators [HAB22], reduced order models [Her20, Men21, HBAB24], matrix-free and High Performance Computing strategies [GHJB25]. Interestingly, several of these dedicated solvers directly employ spline-based geometric models [Wee22, MB24], and even spline composition [HAB22, HBAB24, GHJB25]. Finally, let us notice that a few attempts have been made at shape optimization of geometry-conforming lattices involving full fine-scale simulations, see for instance the optimization of extrusion dies [ZEE22] or microstructured blades [ABB*25]. Yet, the latter does not rely on dedicated solvers of the type mentioned above, and therefore remains limited in scope due to the heavy computational burden. In

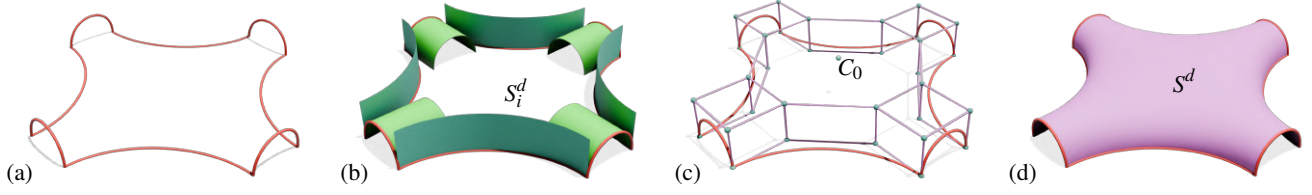


Figure 5: Construction of Generalized Bézier surface. Starting from n (2D or 3D) boundary curves (a), a set of ruled local Bézier surfaces S_i^d (aka ribbons) is built (b). This entails a control structure (c) made up of the ribbon's points and a central control point C_0 that covers the weight deficiency. The n -sided surface S^d (d) is generated by Eq. (8).

this work, we convert our GB model into an analysis-suitable representation, enabling full fine-scale numerical simulations of lattices behavior using spline functions (see Section 6). This, in turn, paves the way for integrating the GB model into such fine-scale design–analysis–shape optimization workflows.

3. The Generalized Bézier tile

In this section we propose a construction of canonical tiles via Generalized (aka n -sided) Bézier patches. While there are more options to construct an n -sided Bézier patch [SVSV20b], we follow the parametric construction of [VSK16], which we recall first.

3.1. Generalized Bézier surfaces

The Generalized Bézier patch (Fig. 4) is in essence a blending of n boundary-based local Bézier surfaces called *ribbons*. These ribbons prescribe the G^0 and G^1 behaviour of the patch in each side. Modeling with Generalized Bézier patches is then a side-based modeling paradigm, in which the designer only prescribes the shape and derivatives of the border curves, something specially well suited for the design of canonical tiles, amongst other design challenges.

A patch $S(u, v)$ is generated by mapping the (u, v) -points of an n -sided ($n \geq 3$) regular convex polygonal domain Γ to $\mathbb{R}^3$ using local coordinates with respect to each side of the domain. Consider domain Γ as the regular polygon given by the set of vertices $\Gamma: \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Local coordinates express each point $(u, v) \in \Gamma$ as a *height* distance $h_i(u, v)$ and a *side* distance $s_i(u, v)$ from an individual side γ_i of the domain. To achieve this, one first expresses each point $\mathbf{x}(u, v) \in \Gamma$ as a function of the vertex of Γ using general barycentric coordinates λ_i such that $\mathbf{x} = \sum_{i=1}^n \mathbf{v}_i \lambda_i(u, v)$, and then writes the side and height functions as

$$s_i = \frac{\lambda_i}{\lambda_{i-1} + \lambda_i}, \quad (1)$$

$$h_i = 1 - \lambda_{i-1} - \lambda_i. \quad (2)$$

Each side of the domain Γ is associated with a Bézier curve of degree d , which forms a side of the 3D patch. Additionally, to interpolate adjacent patches with G^1 continuity, the derivative information in each of these sides must be prescribed. This prescription is done via local Bézier patches S_i^d , called *Bézier ribbons*, defined by the control points $\mathbf{C}_{j,k}^{d,i}$, $0 \leq j \leq d$, $0 \leq k < l$, (see Fig. 5) associated

with the i -th side in the multi-sided patch

$$S_i^d(s_i, h_i) = \sum_{j=0}^d \sum_{k=0}^{l-1} \mu_{j,k}^i \mathbf{C}_{j,k}^{d,i} B_{j,k}^d(s_i, h_i) \quad (3)$$

where $B_{j,k}^d$ are the degree d bivariate Bernstein blending functions of (s_i, h_i) , l is the number of layers in the control point structure (i.e. rings of control points) and $\mu_{j,k}^i$ are scalar multipliers which ensure that the patch at side γ_i is solely determined by control points $\mathbf{C}_{j,k}^{d,i}$. The weights $\mu_{j,k}^i$ are assigned according to Eq. 4 for the $k < 2$ case and Eq. 5 for the $k \geq 2$ case.

$$\mu_{j,k}^i = \begin{cases} \frac{h_{i-1}}{h_{i-1} + h_i} & j < 2 \\ 1 & 2 \leq j \leq d-2 \\ \frac{h_{i+1}}{h_{i+1} + h_i} & j > d-2 \end{cases} \quad (4)$$

$$\mu_{j,k}^i = \begin{cases} 0 & j < k \\ 1/2 & j = k \\ 1 & j > k \end{cases} \quad \mu_{j,k}^i = \begin{cases} 0 & j > d-k \\ 1/2 & j = d-k \\ 1 & j < d-k \end{cases} \quad (5)$$

The shared control points between adjacent ribbons are assumed to coincide exactly, i.e. the ribbons have twist compatibility. To ensure affine invariance, the sum of the blending functions must be one and the construction using ribbons so far does not provide this condition automatically. To account for this deficiency, a central control point $\mathbf{C}_0$ (Fig. 5 (c)) is introduced and multiplied by a central blending function B_0^d that corresponds to the weight deficiency of all the blending functions assigned to the sides of the patch. The central point's contribution is then written as

$$S_0^d = \mathbf{C}_0 B_0^d, \quad (6)$$

where

$$B_0^d(u, v) = 1 - \sum_{i=1}^n \sum_{j=0}^d \sum_{k=0}^{l-1} \mu_{j,k}^i B_{j,k}^d(s_i, h_i). \quad (7)$$

This central control point can be prescribed by the designer or calculated automatically (e.g. as the spatial average of the innermost layer). Finally, the Generalized Bézier patch S^d is given as:

$$S^d(u, v) = \sum_{i=1}^n S_i^d(s_i, h_i) + S_0^d(u, v). \quad (8)$$

For more details about n -sided Bézier patches and their properties, we refer the reader to [VSK16].

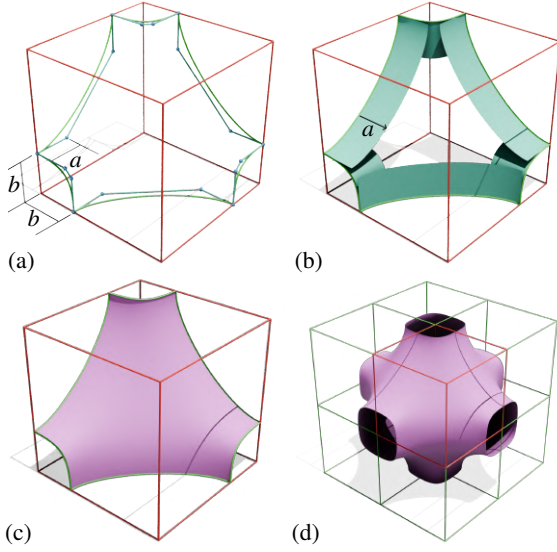


Figure 6: Design of a tile based on a six-sided surface is controlled by two shape parameters a and b . (a) the boundary curves and their control polygons; (b) the Bézier ribbons built from the boundary curves, complying with twist-compatibility; (c) the six-sided surface serves as the base of the tile, embedded in one octant of a cube; (d) the full tile recovered by symmetry.

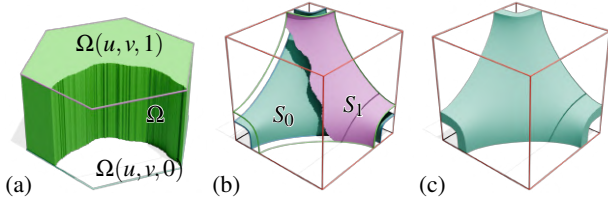


Figure 7: Volumetric parameterization of the shell $V : \Omega \rightarrow \mathbb{R}^3$. (a) shows the parametric hexagonal prism Ω with the top and bottom faces $\Omega(u, v, 1)$ and $\Omega(u, v, 0)$, respectively; (b) shows the Generalized Bézier surfaces $S_0 : \Omega(u, v, 0) \rightarrow \mathbb{R}^3$ and $S_1 : \Omega(u, v, 1) \rightarrow \mathbb{R}^3$; (c) shows the resulting volumetric shell, parametrized via Eq. (9).

3.2. Design of the tiles

To design the tile inscribed in a unit cube, we exploit the symmetry of the tile in this canonical position and model only its $1/8$ (i.e., a part in one octant of the cube). The rest of the geometry is recovered by three consecutive rigid transformations (mirrorings w.r.t the three symmetry planes), see Fig. 6. Each $1/8$ th of the tile is a thick shell constructed from a Generalized Bézier n -sided surface S . To build S in the unit cube we first define positional and derivative information (i.e. the Bézier ribbons) of the boundaries. This is done using cubic Bézier curves on each face of the cube using two parameters a and b , see Fig. 6(a) and (b). Parameter b defines the position of each curve's first and last control points and parameter a defines the position of the interior control points.

Remark 1 In our design of canonical tiles, we assume that the tiles intersect orthogonally the symmetry planes and cell faces, recall

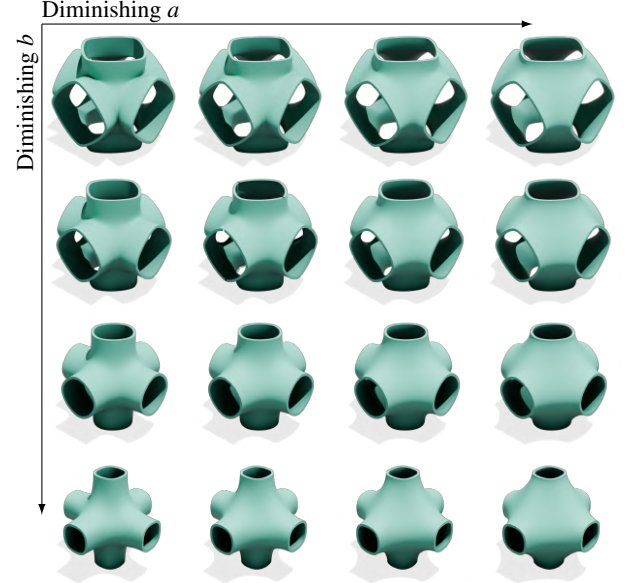


Figure 8: The effect of the parameters a and b on the shape of the tile. Parameter b controls the size of the boundaries of the tile, while parameter a relates to its internal curvature. For this catalogue of shapes the thickness is kept constant.

Fig. 6. This is a sufficient condition to guarantee G^1 continuity between canonical tiles. One could, possibly, relax this orthogonality condition and look for more general setup with a general (non-planar) normal field along the common boundary curves. However, such an approach would increase unnecessarily the complexity in the design of the canonical tiles. As will be seen later in the paper (see e.g. Fig. 12), the orthogonality constraint is relaxed via the global mapping applied to the canonical tiles when constructing the whole lattice.

Once the curves are built, we construct the boundary Bézier ribbons by translating the control polygon of each curve in the direction orthogonal to the cube's face by a distance a . This distance a ensures the correct cross-derivative compatibility between adjacent Bézier ribbons and the orthogonality condition guarantees that each part of the tile – by construction – connects with its symmetrical counterparts, as well as with other neighboring tiles, with at least G^1 continuity[†]. The GB patch S is then built via Eq. (8). The two real numbers a and b are two intuitive shape parameters to control the shape of the GB patch, and consequently the whole tile, see Fig. 8.

To build a thick shell V from this GB patch S we calculate the offset surface S_{off} by a distance t from the original GB patch. This could be done either by an explicit parameterization of the offset

[†] One could, eventually, consider even curvature continuity, but for that one would need to use one extra layer, which would naturally increase the computational cost of the whole construction. Therefore we target only G^1 tiles and G^1 connections.

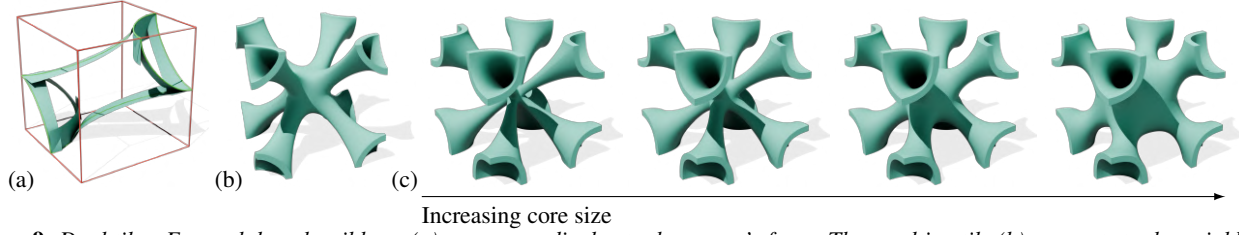


Figure 9: Dual tiles. For each leg, the ribbons (a) are perpendicular to the octant's faces. The resulting tile (b) connects to the neighboring tiles through the vertices of the unit cube instead of the faces. (c) Increasing the core size of the tile by increasing the b parameter for the inside of each leg results in additional shape freedom for the dual tile modeling.

patch, i.e., by symbolic computation of the normals of the original patch, and translating it in the normal direction. However, such an approach would result in a non-rational parametrization of the offset (normalization of normal vector) which is known to be unhandy. Instead, we opt for an offset approximation via calculating the offset of the boundary curves and rebuilding the patch as a GB surface itself. Using both surfaces we produce a volumetric parameterization of $V : \Omega \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ equivalent to a tensor-product Generalized Bézier volume with two layers of control points [QLD24]. In practice this means that the parametric domain of the shell is a 6-sided parametric prism Ω whose top and bottom faces, $\Omega(u, v, 0)$ and $\Omega(u, v, 1)$, represent the parametric domain of the original GB surface and its offset, respectively, see Fig. 7. Interior points are then evaluated as a linear interpolation of their top and bottom faces counterparts:

$$V(u, v, w) = wS_{\text{off}}(u, v) + (1 - w)S(u, v). \quad (9)$$

With the volumetric thick shell at hand, we recover the entire tile just by rigid transformations of this shell. See Fig. 8 for a tile set made up of tiles with different parameters a , b , which gives a clear geometrical intuition for the effect of these parameters on the shape of the tile. Furthermore, if more shape control is required, a third shape parameter can be added by splitting parameter a into external and internal, i.e., one length for the ribbons located on the internal faces of the tile (those on the planes of symmetry) and other for those on the external faces of tile (recall Fig. 6(c)). This has a considerable effect on the shape of the tile, as shown in Fig. 10.

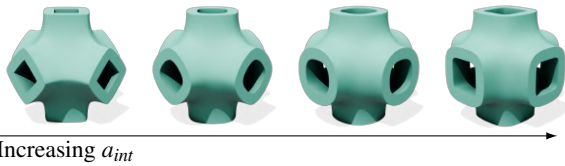


Figure 10: Example of the effect of separating a values for internal (a_{int}) and external ribbons (a_{ext}). All tiles shown have constant a_{ext} and b , while the internal a_{int} varies.

3.3. Dual tiles

Our pipeline was introduced on a canonical tile that was connected to its neighbors via the faces of the surrounding cube, recall Fig. 6. Other possible tile design connects with neighboring tiles via the vertices of the cube, see Fig. 9 (cf. Fig. 6). We call this the *dual*

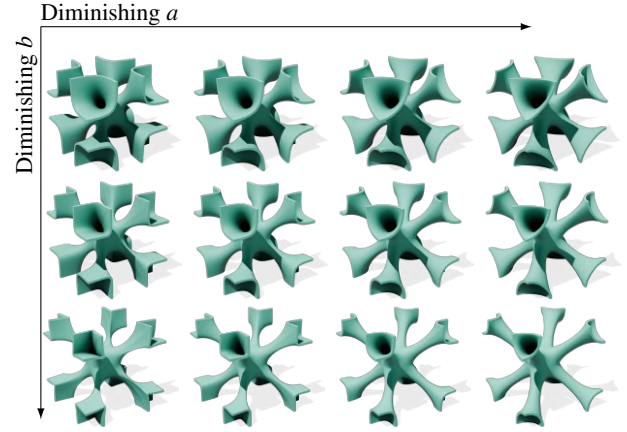


Figure 11: Parameters a and b to control the shape of the tile. Parameter b relates to the size of the boundaries of the tile, while a relates to its internal curvature. In this catalogue the core shape (i.e., the interior a and b) and the tile's thickness t are kept constant.

or *octapod* tile. Such a construction again exploits the symmetry and the leg designed in one octant can be modeled as a pair of 6-sided GB patches, joined again in G^1 fashion with their ribbons orthogonal to the cube's faces. Similarly to the design shown in Fig. 6, the leg that connects the center of the cube and its vertex is designed the way that the ribbons are perpendicular the cube's faces and symmetry planes, see Fig. 9(a). As in the previous tile, parameter a controls the size of the ribbons and b their position with respect to the cube's vertex. Notice, however, that a and b values (or even the thickness value) need not to be equal for the two ends of each leg, thus we are able to prescribe one value for the internal end of the leg (called *core*) and another one for the other end. Fig. 9(b) shows a tile with such property. A gallery of the dual tiles for various values of the external shape parameters a and b is shown in Fig. 11, while the effect of variation of the internal a and b parameters is shown Fig. 9(c).

4. Functional composition using GB tiles

As in previous B-Spline approaches [Elb23], it is also possible to perform functional composition using our GB tiles. Consider a tile defined by some shell $V : P \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and a trivariate B-Spline or Bézier mapping $F : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$, the functional composition



Figure 12: Functional composition with the Generalized Bézier tiles for various geometries. The resulting lattices are of interest for fields such as art, engineering or architecture. The global G^1 continuity is not only aesthetically desirable but also has mechanical implications.

$\zeta : P \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is given by $\zeta(u, v, w) = F(V(u, v, w))$. In practical terms, by filling domain D with $m \times n \times k$ tiles and mapping them via the trivariate map F , one populates the volume defined by F with the tiles, see Fig. 12. Notice that, in essence, the Generalized Bézier construction is fully defined by its Bézier ribbons, therefore the algorithms known for polynomial composition [DGHM93, Elb16] apply in this case as well.

We would like to emphasize that, by construction, the GB tiles are G^1 everywhere and their connections with neighboring tiles are G^1 as well, given that the boundary Bézier ribbons are always orthogonal to the cube faces, and common faces between two adjacent tiles share the local a , b , and t parameters. One could achieve the G^1 continuity without the orthogonality constraints, however, such an approach would require incorporating the G^1 continuity constraint in every cube face (and/or symmetry plane) which would be implementation-wise a lot more complex. In contrast, our approach offers a simple design pipeline that is based on repetitive application of the canonical tile. Note that a globally smooth lattice metamaterial is desirable, e.g., to avoid stress concentrations due to sharp corners and edges [PRZ17]. This requires a globally smooth mapping ζ , so if this natural assumption is met, our pipeline guarantees (at least) G^1 continuity everywhere.

5. Graded domains

Looking towards the optimization of lattices in the physical space, we must be able to smoothly vary the shape of the lattice tiles in

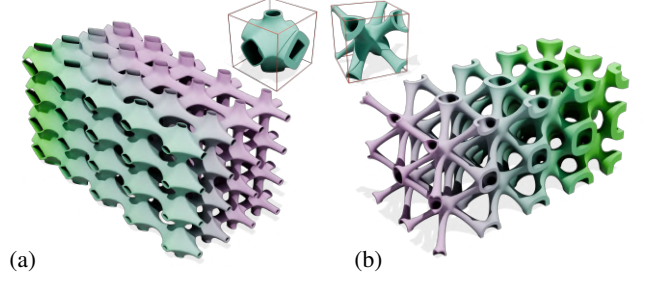


Figure 13: Anisotropic tiling for both cross (a) and dual (b) topologies. The prescription of our shape parameters (and thickness) in a per-tile basis instead of for the whole tile enables anisotropic grading. The canonical tiles are shown (top, inscribed in the unit box).

the parametric space. To do this we consider the shape parameters a , b and the thickness t locally, that is, per face of the tile and not globally as before. We call this construction *graded tiling* and it results in anisotropic tiles as the one shown in Fig. 13-top-framed. Observe that by using this per-face prescription we can generate domains where both the shape, governed by parameters a and b , and thickness, given by t , are graded in some direction while keeping everything in the desired G^1 configuration, see Fig. 13(a) and (b). These graded tiles can then be deformed to achieve graded lattices, both in shape and thickness, which allows the designer to control mechanical response locally throughout the design domain, see Fig. 14 for some examples. Notice that the tiles are thicker and bigger near the supports of the shown structures and get thinner and smaller as the structure evolves away from their supporting region. We demonstrate the mechanical advantages of the graded tiling in the next sections.

6. Towards the design of mechanically relevant lattice structures

Beyond its geometrical versatility, the proposed GB-based model can be readily converted into an analysis-suitable representation, thereby enabling its seamless integration into a design–analysis–optimization workflow aimed at creating novel mechanically effective lattices. In this section, we focus on the volumetric cross-tile geometry, recall Fig. 6(d).

It is important to clarify that the notion of optimization considered here refers to sizing and shape optimization, rather than topology optimization. The objective is to illustrate how a given pre-design can be modified by tuning tile thicknesses as well as local and global geometrical features. This section is prospective: rather than addressing a fully defined optimization problem, we aim to show that the proposed framework enables a flexible control of geometry, which could be leveraged for the future design of metamaterials with enhanced mechanical performance.

6.1. Conversion of the GB model into an IGA-suitable model

We start by converting the GB model into an analysis-suitable form, i.e., a representation expressed in terms of basis functions directly compatible with numerical simulation frameworks. Since our model is built upon spline functions, it is desired to

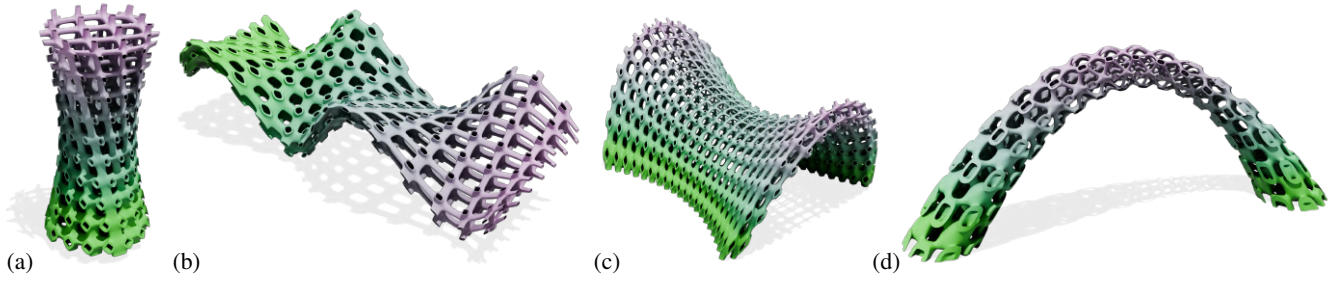


Figure 14: The graded tiles open the door towards optimized and anisotropic lattices. This possibility is important when considering the mechanical behaviour of lattice domains in engineering and architecture.

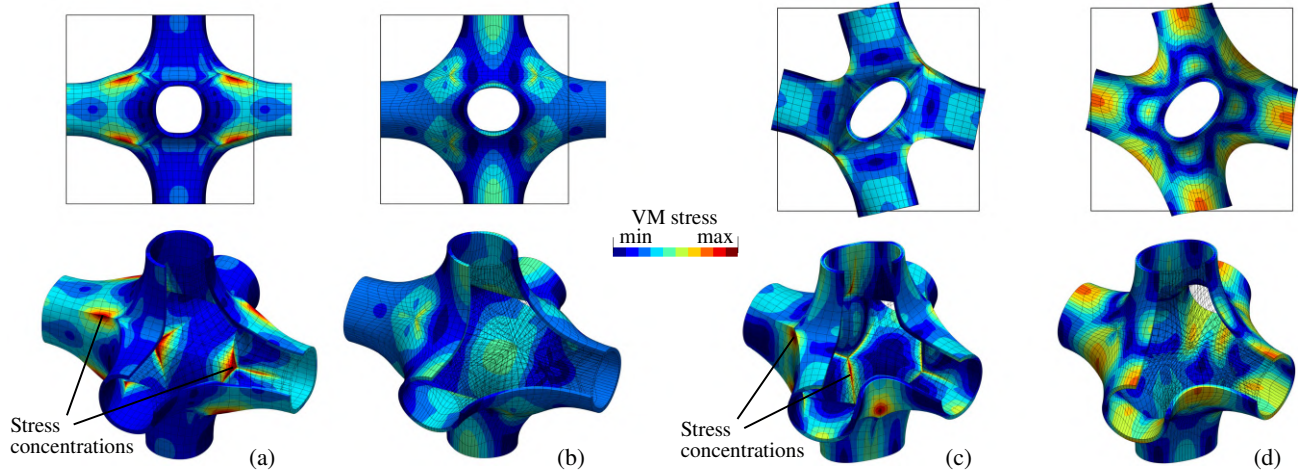


Figure 15: Stress distribution (von Mises stress) in state-of-the-art G^0 (a,c) and proposed G^1 (b,d) cross-tile unit cells under mechanical loading: (a,b) horizontal tension and (c,d) shear. The G^0 tile exhibits pronounced stress concentrations at geometric junctions, both on the exterior and interior of the cell, whereas the G^1 tile provides a smooth stress field free of artificial peaks.

also exploit the advantages of this function family for numerical simulation. Therefore, we seek to convert the GB model into a form suitable for performing the so-called IsoGeometric Analysis (IGA) [HCB05, CHB09, BH22]. In its most widely adopted and robust formulation, IGA requires complex geometries to be represented by conforming tensor-product spline patches, i.e., patches that share control points along their common interfaces. Our objective here is thus to recast the GB model into such a representation, while retaining both an accurate geometric approximation and a high degree of regularity in the model. We do this by an existing subdivision algorithm [YNP97], see Appendix.

Ultimately, the resulting subdivided, tensor-product representation of the GB model is constructed from trivariate Bernstein polynomials, which are associated with multiple conforming patches. This representation can be refined while preserving the geometry using standard spline refinement techniques (knot insertion and degree elevation), providing a rich and flexible function space suitable for solving physical problems with IGA, such as solid mechanics. Specifically, to solve a physical problem on a given tile, one first formulates the weak form of the governing equations and then applies a Galerkin approach using the refined spline space associated with the tile as the approximation space. This method can

also be extended to simulate the behavior of complete lattices, i.e., composed of multiple, potentially graded, cells obtained by spline composition [ABC*19, HBD*19, HAB22, HBAB24]. In the following, we adopt this strategy to simulate the mechanical response of lattices designed using our GB-based approach.

6.2. Impact of G^1 -continuity on structural performance

We aim here to demonstrate the mechanical benefits of the G^1 regularity of our model. For this purpose, two cross-tile unit cells similar in geometry and in mesh refinement are generated: one based on our G^1 formulation and the other on the state-of-the-art G^0 geometry-conforming approach from [Elb16, Elb23]. Both approaches employ a polynomial degree of three and the results are compared in Fig. 15.

For each configuration, two IGA simulations are performed: one where the cells are subjected to a normal macro-strain (see Fig. 15(a,b)) and the other where they are subjected to a shear macro-strain (see Fig. 15(c,d)). The corresponding results are shown in terms of the deformed configuration, with von Mises stress fields represented as contour plots. It can be underlined that, in the G^0 case, the lack of surface smoothness induces artificial stress concentrations at the junctions, which compromise structural

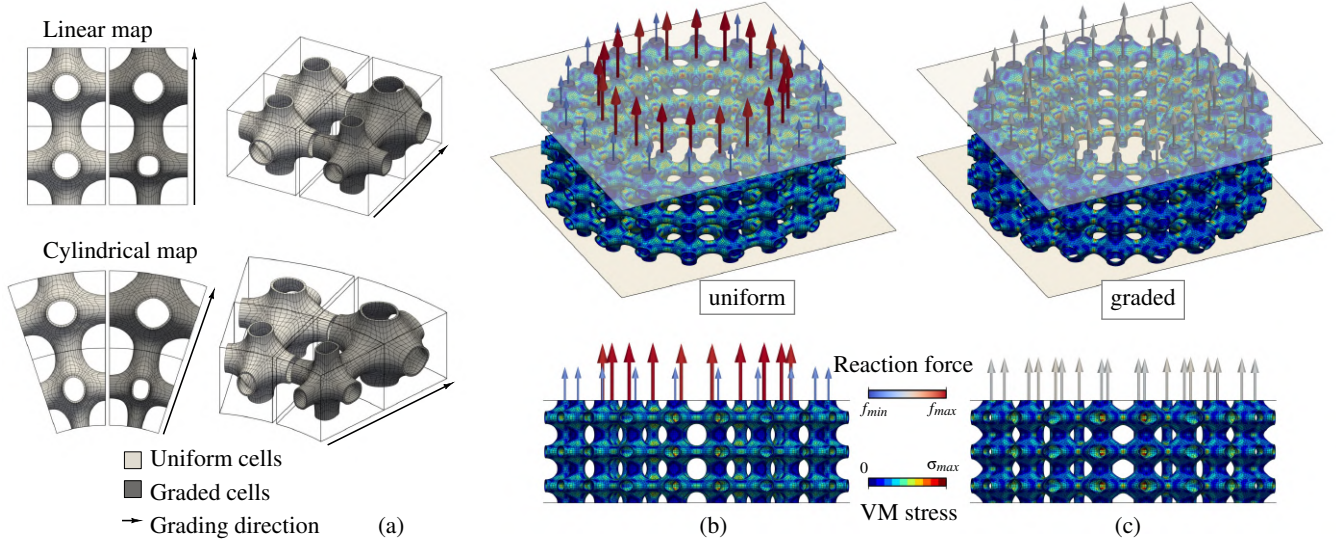


Figure 16: Illustration of graded cylindrical lattice structures: the shape grading can lead to a more uniform reaction force distribution when subjected to compression. (a) shows the involved grading, and (b) and (c) the von Mises stress fields and the top reaction forces (visualized as arrows) for all cells of the uniform and graded lattices, respectively.

strength and integrity. By contrast, the G^1 cell exhibits a smooth stress distribution without localized peaks. This improvement is particularly relevant for design and optimization in a mechanical context, since local stress concentrations inevitably lead to premature structural degradation (see, e.g., [AJM04b, PRZ17]).

Remark 2 Since the structures are subjected to a prescribed level of strains, the average stress level in Fig. 15 characterizes the stiffness of the cell. In particular, the high average stress under shear for the G^1 cell depicted in Fig. 15(d) does not mean that it is weaker than the G^0 cell (Fig. 15(c)). It actually means that it is stiffer, and this is confirmed by the analysis presented in Appendix. In fact, the stress fields in Fig. 15 suggest that the G^0 cell might be, in general, weaker due to the presence of stress concentrations.

6.3. Generating a complete graded lattice structure for potential mechanical applications

Let us now demonstrate the potential of our approach for the design of graded lattice structures with respect to a prescribed mechanical criterion. In particular, we consider a lattice structure with a cylindrical overall shape, a configuration commonly found, e.g., in the design of absorbers for aerospace applications [LSBM23, NDL*23]. The objective here is, for a given unit-cell topology and a prescribed macroscopic shape, to determine an appropriate grading that may help yield a lattice structure capable of distributing the applied load uniformly over its outer surface under compression. This is mechanically relevant, as it may help prevent localized degradation of the support component in contact with the lattice due to excessive reaction forces.

More specifically, the considered structure consists of two-by-twenty-by-three cells in the radial, circumferential, and vertical directions, respectively. Linear elastic IGA simulations are performed at this stage to evaluate its response. The results are presented in

Fig. 16. When only composition is applied without grading, the reaction forces per cell are not balanced, see Fig. 16(b). The inner cells exhibit more reaction forces than the outer cells. In contrast, Fig. 16(c) suggests that, by introducing radial grading, both in shape (through parameters a and b) and thickness, it is possible to reduce these discrepancies and harmonize the reaction forces across all cells. Note that a small thickness grading was needed to reduce the reaction force variations below 2%, where the uniform case exhibits more than 25% variations. This first application illustrates the flexibility and versatility of the proposed methodology in generating novel, graded lattice structures that could be of interest from a mechanical point of view.

6.4. Functional composition to influence mechanical behavior

Finally, we show that functional composition can influence the mechanical behavior of complete lattice metamaterials. For this purpose, we define two test cases as depicted in Fig. 17.

The case of the bending beam suggests that introducing an appropriate smooth and curved macro-mapping (Fig. 17(b)) may lead to a stiffer lattice than the initial straight design (Fig. 17(a)). Indeed, the maximal deflection (or displacement) is reduced by 25%. This indicates potential for design optimization, as also discussed in [ZEE22, ABB*25].

The second test case of a half annulus further illustrates the potential benefit of using geometry-conforming lattices via functional composition instead of clipping (recall Fig. 2). If one tiles the macro-domain with the unit cell, without functional composition, then one may obtain a lattice with less favorable mechanical response (see Fig. 17(c)). In contrast, the functional-composition-based lattice shown in Fig. 17(d) achieves a 25% increase in stiffness, as evidenced by the reduction in maximal displacement.

Remark 3 The simulations presented in this section were imple-

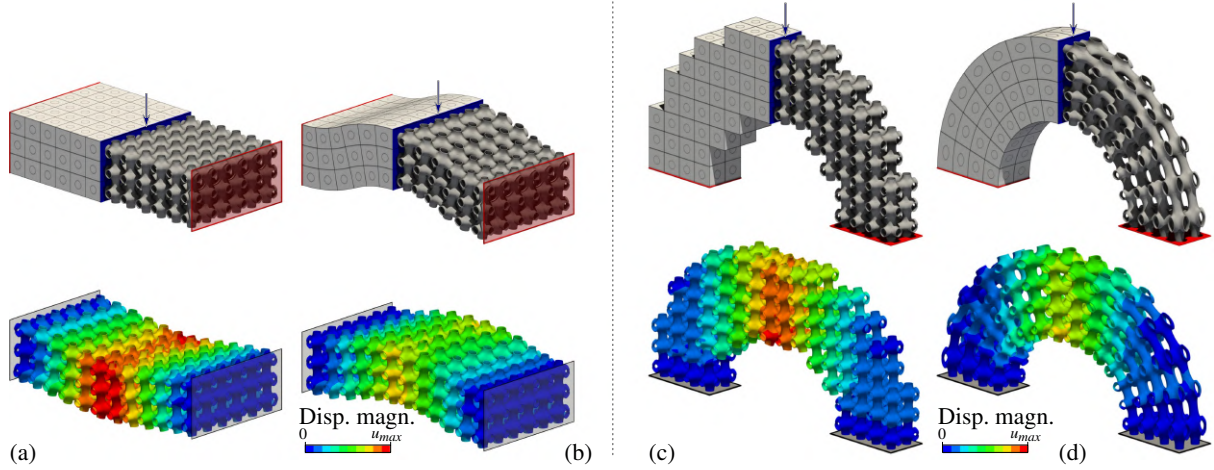


Figure 17: Study of the influence of the macro-mapping on the mechanical behavior of a lattice. (a) and (b) consider a beam under bending with straight and curved macro-mapping, respectively. (c) and (d) consider a half annulus under vertical load with standard clipped and functional composition configurations, respectively.

mented based on a Python library YETI [DHCF*23], and were performed on a standard laptop. The simulation times are in the order of a few minutes on a single core.

7. Discussion & Limitations

Degree of freedom comparison. To showcase the efficiency of our tile representation, we perform a comparison between a state-of-the-art quasi- G^1 B-Spline trivariate tile and our G^1 tile describing approximately the same tile shape, see Fig. 18. The trivariate-based tile is composed of 24 shells, where each shell is linear in the thickness direction and cubic in the other two. This amounts to a total of 768 control points. In contrast, the GB-based tile is composed of 8 shells, where each shell is also linear in the thickness direction and the top and bottom surfaces are six-sided GB patches with cubic ribbons. The GB-based tile leads to a lot more compact geometry representation, needing in total only 384 control points, almost half than the tensor-product variant.

Implicits approximation. Another interesting feature of our tiles is that they can easily approximate triply periodic minimal surfaces, which are currently of great importance in the design and analysis of lattice materials [MB24], using only the few geometric parameters of our model, i.e. a and b . In the present experiment we show the approximation of the Schwarz-P cell, but eventually other implicit tiles could be modeled in the same spirit, e.g. the IWP implicit cell using our dual tile topology. In Fig. 19(a) we show a Schwarz-P benchmark surface retrieved via marching cubes algorithm from its implicit representation, while Fig. 19(b) shows the approximation using eight GB patches, one in each octant, and the approximation error as a percentage of the unit cell (Fig. 19(c)). Observe our construction represents the Schwarz-P geometry with very high accuracy. This result is important as implicits are still difficult to process and simulate efficiently. Therefore, being able

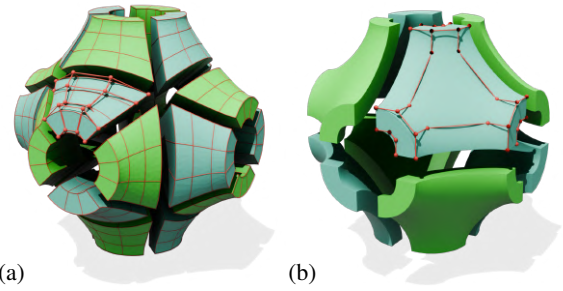


Figure 18: (a) An exploded view of a quasi- G^1 tile using tensor-product trivariates. This tile is composed of 24 trivariates; each trivariate is cubic in two directions and linear in the thickness direction, resulting in 768 control points. In contrast, (b) a Generalized Bézier-based tile describing approximately the same geometry is composed of 8 GB shells, with only 384 control points.

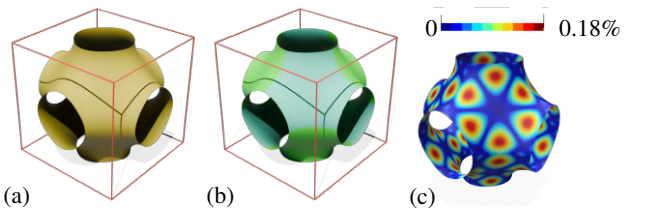


Figure 19: Approximation of an implicit surface with Generalized Bézier. (a) shows a Schwarz-P surface retrieved and (b) its GB-based approximation with eight six-sided patches. Figure (c) shows the error distribution after this approximation.

to approximate and simulate implicit geometries (such as Schwarz-P) using parametric models opens the door to IGA and other well established methodologies. Our parametric surface approximation model is simpler and lighter than those in the literature, e.g. [FJDG-FAC22, ZZL*24].

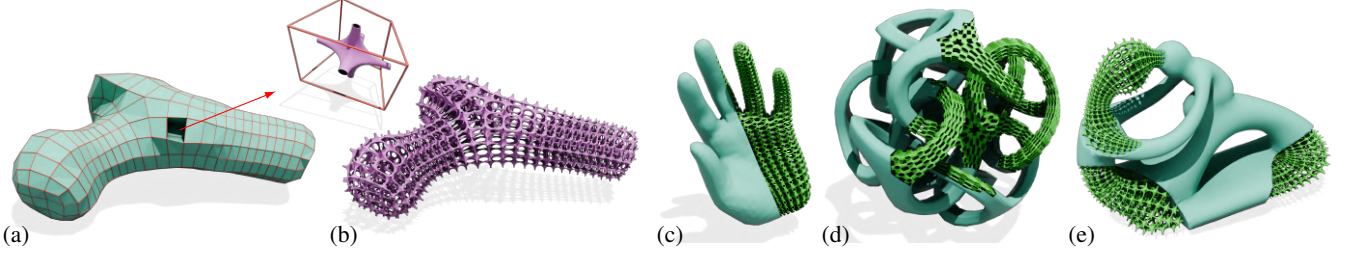


Figure 20: Building lattices out of pure hexahedral meshes. Considering each hexahedral element as a deformation map via a trilinear Bézier (a) we retrieve lattice versions of any pure hexahedral mesh with G^0 connections between adjacent tiles.

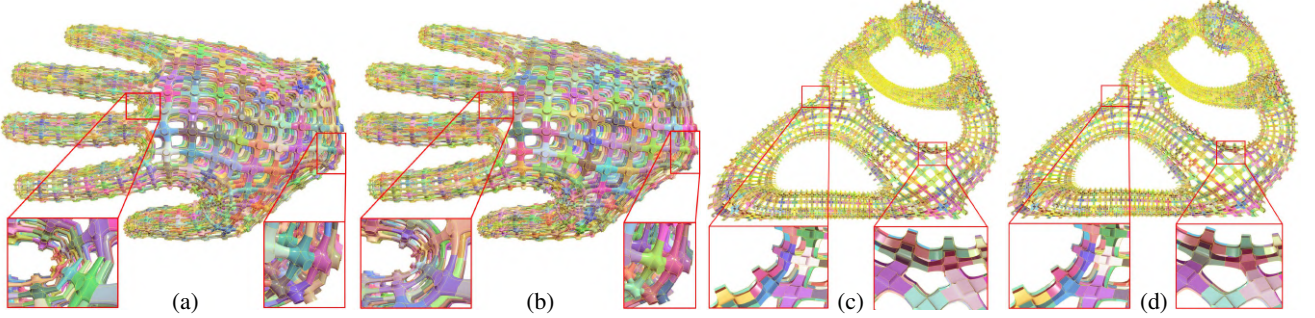


Figure 21: A comparison between C^0 and C^1 lattices of pure hexahedral meshes with cross tiles. (a) and (c) show the C^0 initial results, whereas (b) and (d) present the C^1 results, after averaging the control points on all shared boundaries.

Building lattices out of hexameshes. Other interesting direction is the lattice construction in volumetric domains using not global but local mappings. Given a hexahedral mesh given by $M = (V, H)$, where V are the vertices and H the hexahedral elements, by considering each hexahedral element H_i as a triariate Bézier map $H_i : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, we can deform a tile to match the boundary conditions of the hexahedral element, see Fig. 20(a), and retrieve a lattice version of any pure hexahedral mesh, see Figs. 20(b-e). However, since each cuboid only matches to a trilinear Bézier map, G^1 connection between adjacent tiles is not naturally enforced. This can, however, be fixed in the physical domain; one can think of a construction/optimization that would optimize the layout and the control points of the ribbons such that transition between the tiles would be smoother (e.g. as G^1 as possible in the least square sense). We believe that such a general construction goes beyond the scope of this paper. Yet, for our cross tiles in hand, one could achieve G^1 continuity across tiles, by averaging boundary control points: Assuming the arms of these cross tiles are, say, open-ended and at least tricubic, set all control points of both tiles which are on the boundary between two G^0 hexa domains, to be the average of their respective control points on the 2nd (from boundary) planes, on both sides. The averaged control points from the boundary will not necessarily stay on the boundary (this might be an issue in rare cases) but the result will be a G^1 joint. See Fig. 21 for a few examples.

Initial design in the parametric space. An issue related to shells of constant thickness is the shape control in physical space after the mapping is applied. Our framework designs the canonical tiles in the parametric domain and then maps them to the desired vol-

ume. The deformation function is not an isometry and after the mapping, the geometry (e.g., shell thicknesses) might be modified. Yet, the analysis and optimization are performed in physical space. Hence, the adjustment of the tiles shape parameters (herein a , b and t) following analysis results will consistently converge to the desired outcome. If the application in mind requires lattices with tiles of constant thickness, one could consider the inverse mapping and design the canonical tiles of non-uniform thickness such that deformed tiles would be of constant thickness. At least for shells (thickness negligible with respect to the tile/surface size) this seems doable via the differential of the direct mapping. We did not follow this path as we believe the tiles of non-uniform thickness are more interesting for the majority of the applications as the problems involving physics (loads, stresses) lead to optimal structures with non-uniform thickness.

Geometry conversion to IGA compatible input. The majority of IGA codes require hexahedral-based parametrization of the physical domain to apply tensor-product spline parametrizations. The use of GB patches (and shells) leads towards hexagonal parametric domains (hexagonal prisms in case of shells, recall Fig. 7) which can be converted into the desired IGA-compatible input using a scheme based on [YNP97]. Even though there is some recent works on Generalized Béziers in the context of IGA [WJZ23], we believe that the analysis-suitability of the functional spaces over the hexagonal prisms is a task beyond the scope of the current paper, but is a very interesting direction for future research.

Alternative hole-filling constructions. The construction of [VSK16] is a matter of choice we made. There exist alternative



Figure 22: Fabricated results. Two models from Fig. 12 and Fig. 21 have been 3D printed using Stereolithography (Formlabs Form3+ printer, black resin) and PolyJet (Stratasys J55 printer, multi-material) processes, respectively.

hole-filling constructions that can design a union of patches stick together with G^1 continuity, see e.g. [KP21] and other relevant references cited therein.

Physical feasibility. While we believe a detailed analysis of physical feasibility (e.g. accessibility analysis in case of multi-axis 3D printing, support vs. support-free fabrication, or tile thickness) of the designed geometries goes beyond the scope of the current paper, we 3D printed two specimen to demonstrate their physical feasibility, see Fig. 22.

8. Conclusion and future work

We have proposed a new methodology for the design of globally G^1 geometry-conforming lattice metamaterials using a Generalized Bézier-based canonical tiles. The tiles consist of 6-sided GB shells that are joined in G^1 fashion to produce lattices that are globally G^1 , both in the parametric domain, and via smooth mappings, also in the physical space. The presented tiles are controlled via a set of intuitive shape parameters that can vary across the tiles to form anisotropic structures of graded shape and thickness.

We have demonstrated the proposed pipeline on benchmark geometries and, through the conversion of the GB-model into conforming tensor-product Bézier patches, have shown its compatibility with the analysis framework. In physical-based applications, in particular stress analysis, we have confirmed superior behavior of the proposed G^1 tiles over the conventional G^0 tiling.

Looking ahead, an interesting perspective would be to investigate the possibility of performing IGA directly on the GB-based model. While challenging [WJZ23], such an approach could further increase the power and efficiency of the geometrical modeling framework developed here. Another promising direction lies in the concrete implementation of shape optimization, aiming at the design of mechanically optimal structures. This will, however, likely require the development of dedicated efficient computational strategies [HBA24, GHJB25] to overcome the heavy cost of such simulations.

Acknowledgments

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